Paper: Designing Optimal Binary Rating Systems
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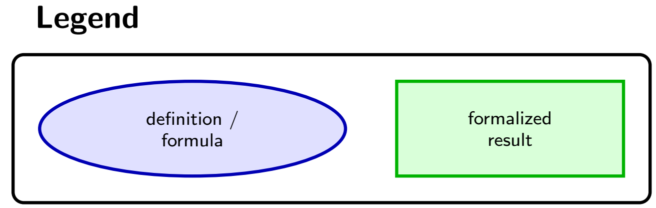

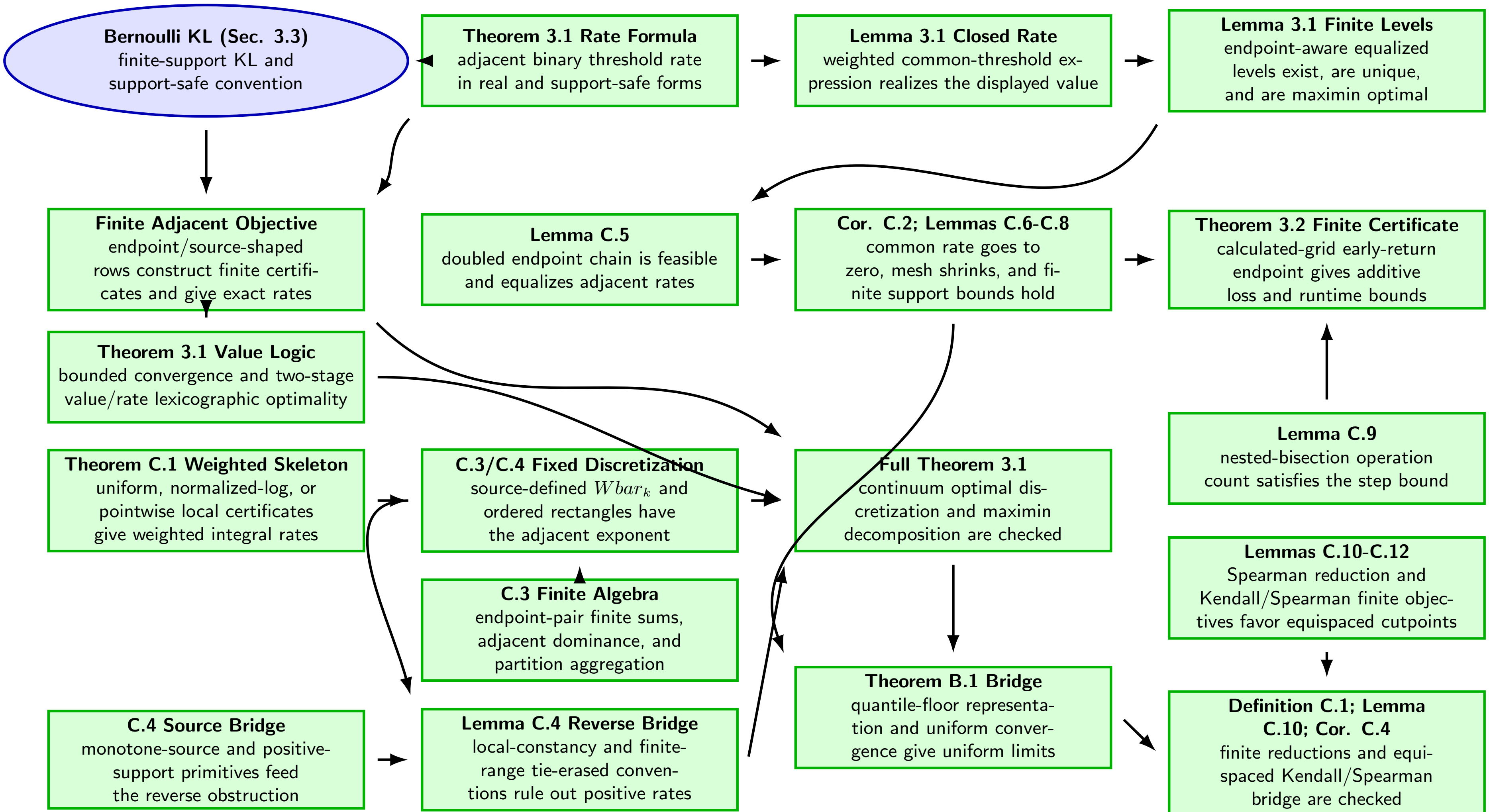

Scope note. Every box in this DAG is a Lean-checked definition, theorem, or result row. Some theorem boxes keep explicit grid, rate, weight, essential-infimum, local pointwise-rate, or iteration hypotheses visible in the paper-facing interface; these are closed theorem statements, not open assumptions or partial nodes. The reusable weighted Laplace skeleton behind Theorem C.1 and the endpoint-pair finite-sum, partition-integral, adjacent-dominance, source-defined fixed-discretization $Wbar_k$, and positive-rate algebra behind Lemmas C.3-C.4 and Theorem 3.1's fixed-discretization bridge are formalized under visible hypotheses. Corollary C.2, the finite Lemma C.10-C.12 example reductions, the reverse C.4 obstruction core, the concrete monotone-interval/local-constancy and finite-range pointwise-LDP reverse bridges, the Theorem 3.2 calculated-grid early-return endpoint, and the common floor-coordinate / canonical B.1/Corollary C.4 convergence branches are formalized. The full Theorem 3.1 continuum discretization path and the Kendall/Spearman convergence examples are closed in the paper-facing interface. The finite Appendix B.2/B.3 learning wrappers are Lean-checked; the empirical/visualization material remains source scope but is not a theorem-DAG target here.